Software Test Plan (STP)

Project: Event Management System (EMS)

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Status: Final demo

1. Introduction

1.1 Purpose

This plan defines the scope, strategy, environment, schedule, responsibilities and test cases for the Event Management System. It traces checks to the functional, non-functional and security requirements in the SRS.

1.2 Scope

Testing covers accounts and authentication, event creation and approval, discovery, registration and waitlist, notifications, check-in, reports, administration, performance, usability, compatibility and security of the EMS web application. Third-party services (SMTP provider, MongoDB internals) are excluded.

1.3 References

EMS SRS v1.0 (requirement IDs); EMS SAD v1.0 (components, API, sequence diagrams); OWASP ASVS 4.0 and Testing Guide; IEEE 829.

1.4 Definitions

SRS, SAD, RTM (Requirements Traceability Matrix), UAT (User Acceptance Testing), TC (Test Case), TLS, RBAC, JWT. Requirement IDs: EMS-F (functional), EMS-NF (non-functional), EMS-SAF (safety), EMS-SEC (security).

2. Test Items

- Auth & User module (C3): registration, login, lockout, password reset, profile, user administration
- Event module (C4): create, edit, approve, cancel, categories, browse and search
- Registration module (C5): registration, seat allocation, waitlist, cancellation
- Notification module (C6): confirmation, reminder, change and decision emails, retry
- Attendance & Reporting module (C7): check-in, dashboard, CSV export, admin summary

- API layer and middleware (C2): authentication, RBAC, validation, rate limiting, error handling
- Web client (C1): main screens, responsive layout, accessibility
- Data layer and database (C8, C9): indexes, backup and restore

3. Features to be Tested

All features are mapped to SRS requirement IDs. Every requirement below has at least one test case (see section 13).

Requirement ID	Feature / requirement (short)	Test case(s)
EMS-F-001	Attendee registration	TC-AUTH-01, TC-AUTH-02
EMS-F-002	Login and session token	TC-AUTH-03
EMS-F-003	Account lockout	TC-AUTH-04
EMS-F-004	Role-based access control	TC-AUTH-05
EMS-F-005	Password reset	TC-AUTH-06
EMS-F-006	Profile update	TC-AUTH-07
EMS-F-007	Create event	TC-EVT-01, TC-EVT-02
EMS-F-008	Event date validation	TC-EVT-02
EMS-F-009	Pending approval status	TC-EVT-01
EMS-F-010	Admin approval / rejection	TC-EVT-03
EMS-F-011	Edit own event	TC-EVT-04
EMS-F-012	Cancel event	TC-EVT-05
EMS-F-013	Browse published events	TC-SRC-01
EMS-F-014	Search and filter	TC-SRC-01
EMS-F-015	Event details	TC-SRC-02
EMS-F-016	Register for event	TC-REG-01, TC-REG-02
EMS-F-017	No duplicate registration	TC-REG-02
EMS-F-018	Waitlist with FIFO promotion	TC-REG-03, TC-REG-04
EMS-F-019	Cancel registration	TC-REG-04
EMS-F-020	My registrations	TC-REG-05
EMS-F-021	Capacity never exceeded	TC-REG-06
EMS-F-022	Registration confirmation email	TC-NTF-01
EMS-F-023	24-hour reminder	TC-NTF-02
EMS-F-024	Change / cancellation / decision notices	TC-EVT-05
EMS-F-025	Retry and delivery log	TC-NTF-03
EMS-F-026	Check-in attendee	TC-CHK-01

Requirement ID	Feature / requirement (short)	Test case(s)
EMS-F-027	Reject invalid check-in	TC-CHK-02
EMS-F-028	Organizer dashboard	TC-RPT-01
EMS-F-029	Export attendee CSV	TC-RPT-01
EMS-F-030	Admin platform summary	TC-RPT-02
EMS-F-031	Manage users and roles	TC-ADM-01
EMS-F-032	Manage categories	TC-ADM-02
EMS-NF-001	Search response time	TC-PERF-01
EMS-NF-002	Registration response time	TC-PERF-02
EMS-NF-003	Availability	TC-AVL-01
EMS-NF-004	Ease of registration	TC-USB-01
EMS-NF-005	Browser and device support	TC-CMP-01
EMS-NF-006	Backup and recovery	TC-AVL-01
EMS-NF-007	Test coverage and code quality	TC-MNT-01
EMS-NF-008	Capacity of the system	TC-PERF-01
EMS-NF-009	Accessibility	TC-CMP-01
EMS-NF-010	Time zone handling	TC-TZ-01
EMS-SAF-001	Confirm destructive actions	TC-SAF-01
EMS-SAF-002	Minimal personal data	TC-SAF-01
EMS-SEC-001	Password hashing	TC-SEC-01
EMS-SEC-002	Encrypted transport	TC-SEC-02
EMS-SEC-003	Server-side authorization	TC-AUTH-05, TC-EVT-04
EMS-SEC-004	Input validation and output encoding	TC-SEC-03
EMS-SEC-005	Rate limiting	TC-AUTH-04, TC-SEC-04
EMS-SEC-006	Audit logging	TC-SEC-05
EMS-SEC-007	Secure session tokens	TC-SEC-06
EMS-SEC-008	Secrets and dependencies	TC-SEC-07

4. Features Not to be Tested

- Internals of MongoDB, Node.js, Express and React libraries (assumed tested by their maintainers).
- Delivery performance of the third-party SMTP service (only EMS behaviour on success and failure is tested; the SMTP service is stubbed or Mailtrap is used).
- Payments, native mobile apps, live streaming, seating charts and multi-language support (out of scope for v1.0, SRS section 1.3).
- Browsers other than the latest two versions of Chrome, Firefox and Edge.

5. Test Approach / Strategy

Levels

Level	Description	Tools	Responsible
Unit	Service-layer functions and validators in isolation (target: 70% statement coverage)	Jest	Developer of the module
Integration	API endpoints with the real database (test DB) and stubbed SMTP	Jest, Supertest, MongoDB test instance	Developer + peer
System	End-to-end flows through the web client; performance, usability and compatibility tests	Manual scripts, Postman, JMeter, Lighthouse	All members (cross-testing)
Acceptance (UAT)	Walk-through of the main flows with the instructor or 3-5 sample users against the SRS	Checklist from section 13	All members

Types

- Functional testing of every FR (black-box, equivalence classes and boundary values, for example capacity 0, 1, 10000, 10001).
- Regression testing: the automated unit and API suites run on every pull request through GitHub Actions.
- Performance testing: the recorded localhost Node HTTP workload substituted for the planned JMeter run with 100 concurrent users.
- Usability and accessibility testing: 5-user task test (EMS-NF-004) and Lighthouse (EMS-NF-009).
- Compatibility testing across browsers and screen widths (EMS-NF-005).
- Security testing as described in section 5.1.

Entry criteria

Stable build on the main branch; test environment and seed data ready; requirement under test is implemented and merged.

Exit criteria

Original exit criteria: execute all planned tests, pass all high-priority cases, meet the coverage target, and resolve critical or high-severity defects. Section 14 records the outstanding results.

5.1 Security Validation

Area	Validation activity	Requirement	Test case
Credential storage	Inspect database, API responses and logs for plaintext passwords; verify bcrypt cost factor	EMS-SEC-001	TC-SEC-01
Transport security	Check HTTP-to-HTTPS redirect and TLS 1.2+ with an SSL scanner	EMS-SEC-002	TC-SEC-02
Authorization	Call every protected endpoint with each role and with another user's resource; expect 403	EMS-SEC-003, EMS-F-004	TC-AUTH-05, TC-EVT-04

Area	Validation activity	Requirement	Test case
Input handling	NoSQL operator injection, XSS payloads, fuzzing of numeric and string fields, oversized bodies	EMS-SEC-004	TC-SEC-03
Brute force and flooding	Exceed login and general rate limits; verify 429 and lockout	EMS-SEC-005, EMS-F-003	TC-SEC-04, TC-AUTH-04
Audit trail	Perform security-relevant actions and verify audit entries and absence of secrets	EMS-SEC-006	TC-SEC-05
Session security	Tampered, expired and deactivated-user tokens; cookie flags HttpOnly, Secure, SameSite	EMS-SEC-007	TC-SEC-06
Secrets and dependencies	Secret scan of repository history; npm audit; check .env not committed	EMS-SEC-008	TC-SEC-07
Penetration testing	Manual scan of the deployed demo with OWASP ZAP (baseline scan) against the Top 10 categories; findings logged as defects	EMS-SEC-003 to 007	Report in Test Summary

Local security evidence is available. Deployed TLS, HTTPS cookie behavior and ZAP remain unexecuted.

6. Test Environment

Hardware	Developer laptops (Windows / macOS / Linux), one cloud or local server for the demo build; Android and iOS phones for responsive checks
Software	Node.js 20, MongoDB 7 (separate test database), EMS web client and API, Chrome, Firefox, Edge
Tools	Jest and Supertest (unit/API), Postman (manual API), JMeter (performance), Lighthouse (accessibility), OWASP ZAP and testssl.sh (security), gitleaks and npm audit (secrets and dependencies), Mailtrap (email), GitHub Issues (defect tracking), GitHub Actions (CI)
Test data	Seed script: 1 Admin, 3 Organizers, 50 Attendees, 10 categories, 5,000 events for load tests, events in every status, registrations in every status

7. Test Schedule

Sprint dates follow the project schedule. Later entries record validation evidence.

Milestone	Date
Test plan and test case design (this document)	05-Oct-2026 to 07-Oct-2026
Test environment and seed data ready	12-Oct-2026
Sprint 1 planned window	19-Oct-2026 to 23-Oct-2026
Sprint 2 planned window	26-Oct-2026 to 30-Oct-2026
Local validation, load and security evidence	02-Nov-2026
Coverage and browser follow-up	08-Nov-2026
Usability evidence merged	10-Nov-2026

8. Test Deliverables

- Test Plan (this document)
- Test cases (section 13), manual and automated
- Test scripts (Jest, Supertest, Postman collection, JMeter plans)
- Test data and seed script
- Test execution logs and CI reports
- Defect reports (GitHub Issues with label 'bug')
- Test Summary Report

9. Roles and Responsibilities

Role	Responsibility
QA Lead (rotating per sprint)	Maintain test plan, coordinate execution, report metrics (Sprint 1; rotates to Member 2, 3, 4 in later sprints)
Test Engineer	Write and execute test cases for own modules and cross-test a teammate's module; log defects
Developer	Fix defects, write unit tests, keep CI green
Product Owner	Approve test results and sign off readiness

Report responsibilities: Prajin coordinated system validation. Palash finalized the unit test report and load review. Rakshan finalized code coverage. Yathindra contributed branch coverage and UI validation.

10. Risks and Mitigation

Risk	Mitigation
Late delivery of stable builds	Merge to main at least twice per sprint; smoke test on every merge via CI
Test environment or database unavailable	Docker Compose or local MongoDB for each member; backup copy of seed data
SMTP limits block email tests	Use Mailtrap sandbox and an SMTP stub; test retries with a failing stub
Concurrency and timing tests are flaky	Use deterministic clock stubs for reminder and token expiry; run concurrency test several times
Team members unavailable during exams	Rotating QA lead; cross-testing so each module has two people who know it

11. Assumptions & Dependencies

- The SRS requirement IDs are stable; changes go through a pull request that updates SRS, SAD and STP together.
- Mailtrap or an equivalent test mailbox is available.
- Seed data and test accounts are provided by the seed script before execution.
- A deployed HTTPS target is required for TC-SEC-02. The localhost demo does not exercise it.

12. Suspension & Resumption Criteria

Suspend testing if: the test environment is unavailable for more than 4 hours; the build is too unstable to run (blocks more than 30% of planned test cases); or a critical defect (data loss, security breach, login impossible) is found.

Resume testing when: blocking defects are fixed and verified by a smoke test; the environment is stable again.

13. Test Case Management & Traceability

The RTM links each SRS requirement to at least one test case and complements SRS Appendix C. Section 13 retains the original procedures and thresholds. Sprint updates and actual evidence are recorded in docs/testing/TEST_CASES_V2.md and the System Validation Report.

13.1 Test cases

ID	Req. IDs	Level / Type	Test case (steps)	Expected result
TC-AUTH-01	EMS-F-001	Integration / Functional	**Register with valid data.** Open Register page. Enter name, unused email, password 'Event123'. Submit.	Account created within 3 s; Attendee role assigned; success message shown; password not shown in response.
TC-AUTH-02	EMS-F-001	Integration / Functional	**Duplicate email and weak password rejected.** Register again with the same email. Then register a new email with password 'abc'.	First attempt rejected with 'Email already in use'; second rejected with password-rule message; no account created.
TC-AUTH-03	EMS-F-002	System / Functional	**Login and token expiry.** Login with valid credentials; note token expiry claim; call a protected API with the token before and after 60 minutes (use a clock stub).	Login succeeds; token expires 60 min after issue; call after expiry returns 401.
TC-AUTH-04	EMS-F-003, EMS-SEC-005	Integration / Functional	**Lockout after 5 failed logins.** Submit a wrong password 5 times for one account, then submit the correct password.	6th attempt refused for 15 minutes with remaining time displayed, even with the correct password; login works after the lock ends.
TC-AUTH-05	EMS-F-004, EMS-SEC-003	Integration / Security	**Role-based access control.** As Attendee call POST /events and PATCH /events/{id}/approval; as Organizer call PATCH /admin/users/{id}.	Every call returns 403 and no data changes.

ID	Req. IDs	Level / Type	Test case (steps)	Expected result
TC-AUTH-06	EMS-F-005	Integration / Functional	**Password reset link is single-use and expires.** Request reset; use the emailed link to set a new password; reuse the same link; separately wait 31 minutes and use a fresh link.	First use succeeds; reuse and expired link rejected; old password no longer works.
TC-AUTH-07	EMS-F-006	Unit / Functional	**Profile update validation.** Update name and phone with 10 digits; then try a 9-digit phone number.	Valid update saved; 9-digit phone rejected with field error.
TC-EVT-01	EMS-F-007, EMS-F-009	System / Functional	**Organizer creates event (pending).** Login as Organizer, submit a valid event form. Then open public listing as a visitor.	Event saved with status Pending Approval; not visible in public listing.
TC-EVT-02	EMS-F-007, EMS-F-008	Unit / Functional	**Event field and date validation.** Submit with: missing venue; end before start; deadline after start; capacity 0; capacity 10001.	Each submission rejected (422) with an error naming the offending field; nothing saved.
TC-EVT-03	EMS-F-010	System / Functional	**Admin approves and rejects events.** As Admin approve event A; reject event B with reason 'too short'(9 chars), then with a 10+ character reason.	A becomes Published and appears publicly; 9-character reason refused; B becomes Rejected; organizer receives both decisions by email.
TC-EVT-04	EMS-F-011, EMS-SEC-003	Integration / Functional	**Edit rules and ownership.** Organizer edits own event capacity from 50 to 5 when 8 are Confirmed; Organizer 2 tries to edit Organizer 1's event.	Capacity reduction rejected; edit of another organizer's event returns 403.
TC-EVT-05	EMS-F-012, EMS-F-024	System / Functional	**Cancel event notifies attendees.** Organizer cancels a Published event that has 3 Confirmed and 2 Waitlisted attendees.	Status becomes Cancelled; 5 cancellation emails queued within 5 minutes; event shows as cancelled to attendees.
TC-SRC-01	EMS-F-013, EMS-F-014	System / Functional	**Browse, search and filter.** Seed 25 published events (some past). Open list; search keyword 'hack'; filter category 'Workshop'; filter date range; tick 'seats available'.	Only upcoming Published events shown, 10 per page sorted by start; each filter returns only matching events; combined filters use AND.
TC-SRC-02	EMS-F-015	Integration / Functional	**Event details and remaining seats.** Open the detail page of an event with capacity 50 and 12 Confirmed registrations.	All required fields shown; remaining seats shown as 38.
TC-REG-01	EMS-F-016	System / Functional	**Register for an event.** Login as Attendee, open a Published event with free seats, click Register.	Confirmed registration created with unique registration ID and QR code; remaining seats reduce by 1.
TC-REG-02	EMS-F-016, EMS-F-017	Integration / Functional	**Duplicate and late registration blocked.** Register twice for the same event; then try to register after the registration deadline.	Second attempt shows 'Already registered' (409); late attempt refused; no extra registrations stored.
TC-REG-03	EMS-F-018	System / Functional	**Full event goes to waitlist.** Fill an event of capacity 2; a third Attendee clicks Register.	Third attendee is Waitlisted with position 1; Confirmed count remains 2.
TC-REG-04	EMS-F-018, EMS-F-019	System / Functional	**Cancellation promotes first waitlisted attendee.** With 2 Confirmed and Waitlist [W1, W2], a Confirmed attendee cancels.	Registration Cancelled; W1 (not W2) promoted to Confirmed; W1 receives promotion email; Confirmed count stays at capacity.

ID	Req. IDs	Level / Type	Test case (steps)	Expected result
TC-REG-05	EMS-F-020	Integration / Functional	**My Registrations list.** Attendee with one Confirmed, one Waitlisted, one Cancelled and one Checked-in registration opens My Registrations.	All four are listed with correct event name, date and status.
TC-REG-06	EMS-F-021	Integration / Functional	**Concurrent registrations never exceed capacity.** Event capacity 10. Script 60 different attendees registering at the same moment (parallel requests).	Exactly 10 Confirmed and 50 Waitlisted; seatsTaken equals 10; no duplicates.
TC-NTF-01	EMS-F-022	System / Functional	**Confirmation email within 60 seconds.** Register for an event using a test mailbox (Mailtrap). Measure time to delivery.	Registration confirmation email is successfully queued/submitted for delivery within 60 seconds of successful registration.
TC-NTF-02	EMS-F-023	Integration / Functional	**24-hour reminder.** Create an event starting in 24 h 5 min with 2 Confirmed attendees; advance the scheduler clock.	Both attendees receive exactly one reminder between 23h45m and 24h15m before start; Waitlisted and Cancelled attendees get none.
TC-NTF-03	EMS-F-025	Unit / Functional	**Failed email retried 3 times and logged.** Stub the SMTP service to fail every call; trigger one confirmation email.	Exactly 3 retries at about 1-minute intervals; log shows status Failed and attempts = 4 (1 initial + 3 retries); app continues to work.
TC-CHK-01	EMS-F-026	System / Functional	**Check-in within the event window. Check-in using the registration ID and QR code succeeds within the event window with timestamps; the attempts 2 h before start and 1 h after end are rejected as outside the window.**	Check-in at 30 min before start succeeds with timestamp; the other two are rejected as outside window.
TC-CHK-02	EMS-F-027	Unit / Functional	**Invalid check-in rejected.** Check in the same registration twice; check in a Waitlisted registration; check in with a registration of another event.	Each attempt is rejected with a message stating the reason; no state change.
TC-RPT-01	EMS-F-028, EMS-F-029	System / Functional	**Organizer dashboard and CSV export.** Event with 8 Confirmed (6 checked-in), 3 Waitlisted, 2 Cancelled. Open dashboard; click Export CSV.	Dashboard shows 8 / 3 / 2 / 6 and attendance 75%; CSV has the required 5 columns and one row per registration.
TC-RPT-02	EMS-F-030	Integration / Functional	**Admin platform summary.** Seed known data (events in each status, users in each role, registrations over 30 days). Open the summary as Admin.	Counts and daily registration series match the seeded data.
TC-ADM-01	EMS-F-031	Integration / Functional	**Deactivate user and change role.** Admin deactivates a logged-in Attendee; that user calls any API and tries to log in again; Admin then changes another user to Organizer.	Deactivated user gets 401 on next request and cannot log in; promoted user can access Organizer endpoints.
TC-ADM-02	EMS-F-032	Integration / Functional	**Manage categories.** Admin adds 'Seminar', renames it, then disables it; Organizer opens the create-event form.	Category appears after adding; disabled category is not selectable for new events but still shown on existing events.

ID	Req. IDs	Level / Type	Test case (steps)	Expected result
TC-PERF-01	EMS-NF-001, EMS-NF-008	System / Non-functional	**Search load test.** Seed 1,000 registered users and 5,000 events. Run a baseline search test and then run JMeter with 100 concurrent users for 5 minutes under the required dataset.	95th percentile response time is at most 2 s, error rate is below 1%, and response-time degradation compared with the baseline does not exceed 10%. Design inspection confirms that the API tier maintains no per-user session state in server memory.
TC-PERF-02	EMS-NF-002	System / Non-functional	**Registration load test.** JMeter: 100 concurrent users register for different events for 5 minutes.	95th percentile response time at most 3 s; error rate below 1%.
TC-AVL-01	EMS-NF-003, EMS-NF-006	System / Non-functional	**Health check, availability and backup restore.** Run an uptime monitor against /health for the test period; run the backup script and restore it into a clean database.	Measured availability at least 99%; restored database passes a record-count and sample-record comparison.
TC-USB-01	EMS-NF-004	System / Non-functional	**Usability test of registration.** Give 5 first-time users the task 'find a workshop and register'. Time them and count clicks.	At least 4 of 5 finish within 3 minutes using no more than 5 clicks without help.
TC-CMP-01	EMS-NF-005, EMS-NF-009	System / Non-functional	**Cross-browser, responsive and accessibility check.** Run the main flows in latest Chrome, Firefox, Edge at 360 px and 1920 px widths; run Lighthouse on 4 key pages.	No functional defect or horizontal scroll; Lighthouse Accessibility at least 90 on each page.
TC-MNT-01	EMS-NF-007	Unit / Non-functional	**Coverage and lint gate.** Run 'npm test -- --coverage' and 'npm run lint' in CI.	Service-layer statement coverage at least 70%; ESLint reports 0 errors.
TC-TZ-01	EMS-NF-010	Unit / Functional	**UTC storage and IST display.** Create an event for 10:00 IST; inspect the stored value; view it in the UI.	Database holds 04:30 UTC; UI shows 10:00 IST.
TC-SAF-01	EMS-SAF-001, EMS-SAF-002	System / Non-functional	**Confirmation dialogs and data minimisation.** Trigger cancel event, cancel registration and deactivate user from the UI; then inspect the users collection and registration form fields.	Each action shows a confirmation naming the item and does nothing until confirmed; only the minimum required personal data is stored, while required security metadata such as user ID and IP address may be retained for audit logging.
TC-SEC-01	EMS-SEC-001	Integration / Security	**Password storage and leakage.** Register a user, inspect the users collection, API responses and application logs.	Only a bcrypt hash (cost at least 10) is stored; plaintext password appears nowhere.
TC-SEC-02	EMS-SEC-002	System / Security	**TLS enforcement.** Call the deployed site over http://; run an SSL scan (for example testssl.sh or SSL Labs).	HTTP redirects to HTTPS; only TLS 1.2 or higher accepted.
TC-SEC-03	EMS-SEC-004	Integration / Security	**Injection and XSS attempts.** Send {"email":{"$ne":null}} to login; enter <script>alert(1)</script> as event title and name; fuzz numeric fields with negative, huge and non-numeric values.	NoSQL operator payload rejected; script text stored as plain text and displayed encoded (no alert); invalid values rejected with 422.
TC-SEC-04	EMS-SEC-005	Integration / Security	**Rate limiting.** Send 11 login requests and 101 other API requests within one minute from one IP.	Requests beyond each limit receive HTTP 429.

ID	Req. IDs	Level / Type	Test case (steps)	Expected result
TC-SEC-05	EMS-SEC-006	Integration / Security	**Audit log content.** Perform login, failed login, role change, event approval, cancellation and check-in; read the audit log.	One entry per action with user ID, action, IP and timestamp; no passwords or tokens present.
TC-SEC-06	EMS-SEC-007	Integration / Security	**Token tampering and cookie flags.** Modify the token payload; use an expired token; use the token of a deactivated user; inspect the cookie flags.	All three tokens rejected with 401; cookie has HttpOnly, Secure and SameSite=Strict.
TC-SEC-07	EMS-SEC-008	System / Security	**Secrets and dependency scan.** Run a secret scan (for example gitleaks) over the repository history; run 'npm audit --omit=dev'.	No secrets found in the repository; zero high or critical vulnerabilities.

13.2 Requirement-to-test mapping (RTM)

Functional requirements

Requirement	Test cases	Levels
EMS-F-001	TC-AUTH-01, TC-AUTH-02	Integration
EMS-F-002	TC-AUTH-03	System
EMS-F-003	TC-AUTH-04	Integration
EMS-F-004	TC-AUTH-05	Integration
EMS-F-005	TC-AUTH-06	Integration
EMS-F-006	TC-AUTH-07	Unit
EMS-F-007	TC-EVT-01, TC-EVT-02	System, Unit
EMS-F-008	TC-EVT-02	Unit
EMS-F-009	TC-EVT-01	System
EMS-F-010	TC-EVT-03	System
EMS-F-011	TC-EVT-04	Integration
EMS-F-012	TC-EVT-05	System
EMS-F-013	TC-SRC-01	System
EMS-F-014	TC-SRC-01	System
EMS-F-015	TC-SRC-02	Integration
EMS-F-016	TC-REG-01, TC-REG-02	System, Integration
EMS-F-017	TC-REG-02	Integration
EMS-F-018	TC-REG-03, TC-REG-04	System
EMS-F-019	TC-REG-04	System
EMS-F-020	TC-REG-05	Integration
EMS-F-021	TC-REG-06	Integration
EMS-F-022	TC-NTF-01	System
EMS-F-023	TC-NTF-02	Integration

Requirement	Test cases	Levels
EMS-F-024	TC-EVT-05	System
EMS-F-025	TC-NTF-03	Unit
EMS-F-026	TC-CHK-01	System
EMS-F-027	TC-CHK-02	Unit
EMS-F-028	TC-RPT-01	System
EMS-F-029	TC-RPT-01	System
EMS-F-030	TC-RPT-02	Integration
EMS-F-031	TC-ADM-01	Integration
EMS-F-032	TC-ADM-02	Integration

Non-functional and safety requirements

Requirement	Test cases	Levels
EMS-NF-001	TC-PERF-01	System
EMS-NF-002	TC-PERF-02	System
EMS-NF-003	TC-AVL-01	System
EMS-NF-004	TC-USB-01	System
EMS-NF-005	TC-CMP-01	System
EMS-NF-006	TC-AVL-01	System
EMS-NF-007	TC-MNT-01	Unit
EMS-NF-008	TC-PERF-01	System
EMS-NF-009	TC-CMP-01	System
EMS-NF-010	TC-TZ-01	Unit
EMS-SAF-001	TC-SAF-01	System
EMS-SAF-002	TC-SAF-01	System

Security requirements

Requirement	Test cases	Levels
EMS-SEC-001	TC-SEC-01	Integration
EMS-SEC-002	TC-SEC-02	System
EMS-SEC-003	TC-AUTH-05, TC-EVT-04	Integration
EMS-SEC-004	TC-SEC-03	Integration
EMS-SEC-005	TC-AUTH-04, TC-SEC-04	Integration
EMS-SEC-006	TC-SEC-05	Integration
EMS-SEC-007	TC-SEC-06	Integration
EMS-SEC-008	TC-SEC-07	System

Summary: 44 test cases (29 functional, 7 non-functional, 8 security) covering 52 of 52 requirements.

14. Test Metrics & Reporting

- Backend Jest rerun: 336 passed. Earlier runs: 54 frontend and 47 real MongoDB checks passed.
- Backend module coverage: 91.87% statements and 82.44% branches.
- Local browser flows passed at both widths. Four Lighthouse accessibility scores were 100.
- Usability ran with five users. Average about 100 s and 4 clicks. Individual time limits unverified.
- Local load substitute: search p95 267.22 ms. The 10% degradation criterion failed.
- Deployed TLS, HTTPS cookie transport and ZAP were unexecuted. Overall validation is partial.

15. Approvals

Role	Name
QA Lead	Rakshan R (PES1UG24CS365)
Dev Lead	Palash Agarwal (PES1UG24CS312), Veladi Yathindra Deveswar ((PES1UG24CS854)
Product Owner/ Product Lead	Prajin Ravi Shankar (PES1UG24CS328)